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COS-AT2

1] Apriori Algorithm - Online Shopping Transactions

Given dataset:

Order ID	Products Purchased
O ₁	mobile, Earphones
O ₂	mobile, charger
O ₃	Earphones, charger
O ₄	mobile, Earphones, charger
O ₅	Mobile, Earphone

Assume minimum support = 40%.

Pseudo Code

ALGORITHM APRIORI

INPUT:

Transaction database D

minimum support threshold minsup

OUTPUT:

frequent itemsets

BEGIN

1. Scan database D.

2. Find the support count of every individual item.

3. Generate candidate 1-itemsets C₁.

4. Prune item whose support is less than minsup.

5. Store remaining items as frequent itemset L₁.

6. Set k = 2

7. WHILE L_(k-1) is not empty DO

Generate C_2 :

b. For each candidate itemset C in C_k Do
Count the number of transactions
Containing C .

c. Calculate support:

$$\text{Support}(C) =$$

$$\text{Support Count}(C) / \text{Total Transactions}$$

d. Remove candidates whose support is less than min_sup

e. Store remaining candidates as L_k .

f. Set $k = k + 1$

8. ENDWHILE

9. RETURN all frequent itemsets.

END

Candidate generation and Pruning

For C_1 :

Item	Support count	Support
Mobile	4	80%
Earphones	3	60%
charger	3	60%

All three satisfy the 40% minimum support, so

$L_1 = \{\text{mobile}\}, \{\text{Earphones}\}, \{\text{charger}\}$

Generate C_2 :

- $\{\text{mobile, Earphones}\} \rightarrow 3 \rightarrow 60\%$
- $\{\text{mobile, charger}\} \rightarrow 2 \rightarrow 40\%$
- $\{\text{Earphones, charger}\} \rightarrow 2 \rightarrow 40\%$

Generate C3:

• {mobile, earphones, charger} $\rightarrow 1 \rightarrow 201$

Since $201 < 401$, it is pruned

2] FP-Growth Algorithm - Library Book Borrowing

Dataset:

Borrow ID	Books Borrowed
B ₁	Data Science, Python
B ₂	Python, Machine Learning
B ₃	Data Science, Machine Learning
B ₄	Data Science, Python, Machine Learning
B ₅	Python, Machine Learning

Pseudo Code

ALGORITHM FP-Growth

Input:

Transaction database D
minimum support minsop

Output:

Frequent itemsets

BEGIN

1. Scan D and count the frequency of every item
2. Remove items whose support is less than minsop
3. Sort remaining items in descending order of frequency
4. Create an empty FP-Tree
5. For each transaction T in D DO
 - a. Remove infrequent items from T

- b. Sort items in τ according to global frequency order
- c. Insert τ into the FP-Tree
- d. Increase the count of existing nodes or create new nodes.
6. Create a header Table to connect identical items in the FP-Tree
7. Combine the generated itemsets.
8. Return all frequent itemsets
- END

FP-Tree Construction

First Count the items:

Item	Count
Python	4
ML	4
DS	3

Transactions are reordered before insertion:

- B₁ - Python, DataScience
- B₂ - Python, Machine Learning
- B₃ → Machine Learning, DataScience
- B₄ → Python, machine Learning, DataScience
- B₅ → Python, Machine Learning

The common prefix share nodes in the FP-Tree. This reduces repeated storage.

3] Association Rule Generation

Given

Service Combination	Support count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Formula

$$x \rightarrow y$$

Confidence is:

$$\text{Confidence}(x \rightarrow y) = \frac{\text{Support}(x \cup y)}{\text{Support}(x)}$$

Assume minimum Confidence = 70%.

Pseudo code

ALGORITHM Generate Association Rules

Input:

frequent itemsets F
Support counts
minimum Confidence minconf

Output

Strong association rules

BEGIN

- for each frequent itemset I in F DO
- Generate all non-empty Proper subsets of I

4. $y = I - x$

5. Generate rule $x \rightarrow y$

6. Add rule to the result set

7. END IF

8. END for

9. End for

10. Return all strong association rules

END

Example

{Internet, Streaming}

Support count = 3

Rule

Internet $\rightarrow$ Streaming

$$\text{Confidence} = \frac{\text{support}(\text{Internet, Streaming})}{\text{support}(\text{Internet})}$$

$$= \frac{3}{4} = 75\%$$

Since

$$75\% > 70\%$$

the rule is accepted

Streaming $\rightarrow$ Internet $\Rightarrow 70\%$.

Streaming $\rightarrow$ Cloud storage = 50%.

Since $50\% < 70\%$, the rule is pruned

4. Constraint-Based Frequent Itemset Mining

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Constraint:- Only itemsets containing Python are considered

Pseudocode

Algorithm: Constraint-Based Frequent Itemset Mining

Input:

Student course transaction dataset

minimum support threshold

Required course = "Python"

Output:

Frequent itemsets containing Python

BEGIN

1. Read all student transactions
2. Find the support of Python
3. If Python is frequent: select {Python}
4. Generate itemsets with Python:
 - {Python, Database}
 - {Python, AI}
 - {Python, Database, AI}
5. For each itemset:
 - check its support.
 - If support $\geq$ minimum support:
 - Keep this itemset

else

Remove the itemset

Donot Consider itemsets
without python

Display the frequent
itemsets

STOP

Explanation

The constraint means that Python must be present in every itemset

For example:

- {Python} $\rightarrow$ keep
- {Python, Database} $\rightarrow$ keep
- {Python, AI} $\rightarrow$ keep
- {Database, AI} $\rightarrow$ Remove because Python is missing

5. Multilevel Association Rule Mining

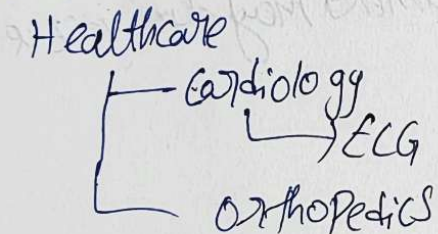
Patient ID	Treatments Received
P ₁	Healthcare, Cardiology, ECG
P ₂	Healthcare, Orthopedics
P ₃	Healthcare, Cardiology
P ₄	Healthcare, Cardiology, ECG
P ₅	Healthcare, Orthopedics

Pseudocode

START

Read all Patient treatment records

Create treatment hierarchy



find frequent treatments at higher level

Generate association rules from frequent treatments

move to the lower level of the hierarchy

find frequent combinations again

Generate rules at each level

Display the association rules

Stop

~~Expm~~

Explanation

multilevel association rule mining finds relationship at different levels of the treatment hierarchy

Level 1:-

- Healthcare

Level 2:

- Healthcare $\rightarrow$ Cardiology
- Healthcare $\rightarrow$ Orthopedics

Level 3:

Cardiology $\rightarrow$ ECG

For example, patients receiving Cardiology treatment may also receive ECG